How the Portfolio Optimizer Works

TradingAgents + the Roth IRA optimizer layer · methods, models and the ChatGPT prompt · 23 October 2026

Overview

The project has two layers. At the bottom is the upstream **TradingAgents** framework (Tauric Research, v0.5.2), a research tool in which a team of LLM agents gives *one ticker on one date* a single rating. On top of it sits a local **portfolio optimizer**, which answers a different question: *what mix should the Roth IRA hold, and what should be bought today?* Neither layer connects to Robinhood or places orders. They only produce suggestions.

Layer 1: TradingAgents (one ticker → one rating)

`TradingAgentsGraph.propagate(ticker, date)` runs a LangGraph pipeline that is set up like a small trading firm:

- **Analysts** (in parallel): market/technicals, sentiment, news and fundamentals. Their data tools are locked to the run's ticker and date, so a past run never sees later data.
- **Bull and bear researchers** debate the findings, and a **Research Manager** sums up.
- A **Trader** writes a proposal.
- **Aggressive, conservative and neutral risk analysts** debate that proposal.
- A **Portfolio Manager** gives the final rating: Buy, Overweight, Hold, Underweight or Sell (REVIEW if no rating can be read).

Memory. Each decision goes into a markdown log. When the same ticker is analysed again later, the old call is scored against SPY, and the LLM writes a short lesson that later runs read. Nothing is fine-tuned.

Data. A vendor router with fallback chains: yfinance, Alpha Vantage, SEC EDGAR, FRED, Polymarket, Reddit and StockTwits.

Layer 2: the portfolio optimizer — methods

Method	Where	What it does
Risk statistics	optimizer/data.py	10 years of daily returns give volatility, covariance (adjusted so the matrix stays valid) and a Sharpe ratio measured against what cash really earned on each day. **Stress loss** per fund = its worst historical drop from peak. Young funds take the crash history of an older stand-in (QQQM→QQQ, AVUV→VBR) or are assigned a 50% floor.
Expected returns with shrinkage	plan.py · expected_excess	Expected return = volatility × a blended Sharpe: each fund's own past Sharpe mixed with a common prior of 0.3, and the less history, the closer it stays to the prior. A $-\frac{1}{2}\cdot vol^2$ term converts it to compound growth, so volatility drag on leveraged funds is priced in.
AI tilt	optimizer/ai_views.py	Runs the Layer 1 agents (market and news analysts) on each holding. Each rating step above or below Hold shifts Sharpe by ±0.15. Ratings change expected returns only, never the loss limit. Same-day ratings are reused from the memory log, so a second run costs nothing.

Method	Where	What it does
Constrained mean-variance solver	optimizer/solve.py	Maximise growth·x − (λ/2)·$x^T\Sigma x$ subject to 0 ≤ x ≤ 35%, Σx ≤ 1 and **Σ weight × stress loss ≤ loss limit**. Solved with an augmented Lagrangian plus accelerated (FISTA) projected gradient, in numpy without scipy. No fund may use more than 25% of the loss budget. Whatever is not invested stays in SGOV.
Screening	universe.py · scan_list	About 570 tickers (ETFs, the S&P 500, small caps, disruptive themes) are scored cheaply, without covariance. The best 60 plus your holdings go to the solver, and funds with under 3 years of prices are skipped.
Loss-limit frontier	plan.py · build_plan	Re-solves the problem at every risk level to show how much growth each step up buys.
T+1 daily orders	plan.py · daily_orders	Buys spend only settled cash. Proceeds from trims are reinvested the next day. New money comes out of SGOV at the daily pace (sold today, settles tomorrow). Trades under 0.5% of the portfolio are skipped. Each order comes with a reason.
Walk-forward backtest	optimizer/backtest.py	Monthly rebalances since 2012, each using only prices known on that date, against VTI, 60/40 and the current mix. It cannot test AI ratings or survivorship; BIL stands in for SGOV before 2020.
Risk levels	optimizer/risk.py	A 1–10 dial mapped to stress-loss caps: 1%, 2.5%, 5%, 10%, 15%, 20%, 30%, 40%, 50%, and none at level 10. Shared by the web app and the CLI through my_risk.json.
Paper tracker	optimizer/tracker.py	Two paper books start from your real holdings, “followed the plan” and “kept as is”. Both are updated after each final close under T+1, and saved to paper_tracker.json.
Market brief	optimizer/market.py	Index, sector and theme returns, S&P 500 breadth, leaders and laggards, plus an optional AI explanation that must cite numbered headlines (otherwise it is marked low confidence).
Web app	webapp/server.py	A local-only page (127.0.0.1): Overview, Performance (since day one, with 1M–All ranges, return, annualized return and drawdown), Allocation, Screener, Backtest and Settings, with the risk-level slider on top.

Models used

The models are set in `tradingagents/default_config.py`. Your `.env` holds API keys only and overrides no model, so the defaults below are the ones that run.

Role	Provider / model	Used by
Deep-thinking model	OpenAI **gpt-6-sol**	Research Manager and Portfolio Manager: the two steps that make the final judgement.
Quick-thinking model	OpenAI **gpt-6-luna**	All four analysts, the bull and bear researchers, the Trader, the three risk debaters, the memory reflections, and the first attempt at the Today-page market brief.
Market-brief fallback	Google **gemini-3.8-flash**, then **gemini-3.5-flash**	Only the Today-page explanation, when the OpenAI call fails (for example, out of credit). It needs GOOGLE_API_KEY, which is set.
No LLM	numpy / pandas	All the optimizer maths: statistics, solver, frontier, orders, backtest and tracker. These give the same result every time and cost nothing.

Changing them. Set `TRADINGAGENTS_LLM_PROVIDER`, `TRADINGAGENTS_DEEP_THINK_LLM` and `TRADINGAGENTS_QUICK_THINK_LLM` in .env. Supported providers include OpenAI, Anthropic, Google, xAI, DeepSeek, Qwen, GLM, MiniMax, OpenRouter, Mistral, Kimi, Groq, NVIDIA, Bedrock, Azure, Ollama and any OpenAI-compatible endpoint.

Current status. The OpenAI key ran out of credit on 22 October (`insufficient_quota`). AI ratings therefore fail until credit is added or the provider is switched. The market brief keeps working through Gemini.

Your inputs and commands

- `my_portfolio.json`: holdings and settled cash. `my_buys.json`: recurring buys (about $27.41 per trading day). `my_universe.json`: which groups to scan (etfs, sp500, themes).
- `.venv/Scripts/python.exe -m cli.main web` opens the web app.
- `.venv/Scripts/python.exe -m cli.main suggest` prints today's orders at the saved risk level; `--risk 5` picks and saves a level; `--ai` adds agent ratings.
- `.venv/Scripts/python.exe -m cli.main track` records the paper portfolios for every closed day; run it daily (e.g. Windows Task Scheduler) so the record never has gaps, even without opening the app.

Things to keep in mind

- **Risk is a 1–10 level, not a fixed 1%.** Level 1 caps the stress loss at 1%, level 9 at 50%, and level 10 has no stress cap (per-fund caps only). On 23 Oct the optimizer invested about 2% at level 1, 29% at level 5 and 98% at level 10. The chosen level is saved in my_risk.json; it starts at level 2 (2.5%), the level the paper record began at.
- The stress loss bounds a *repeat* of history. A future crash can be deeper than any past one.
- **Performance from day one.** The paper record (started 22 Oct 2026) stores, after every close, the value of the suggested portfolio and of your unchanged one, the risk cap in force and the share invested. A risk change never restarts it, and “Start over” archives rather than deletes.

ChatGPT prompt

Paste this into ChatGPT once per trading day, after filling in settled cash and your risk level. It does the same method by hand. The app does the maths exactly, so if the two disagree, trust the app and use ChatGPT as a second opinion. The share counts come from my_portfolio.json as of 23 October 2026; update them when they change.

```
You are my portfolio risk analyst. Each trading day you give me today's orders for my Roth IRA at
Robinhood. Follow every rule below exactly and show your arithmetic.

## My account
- Roth IRA, a cash account. Selling is not taxed.
- Settlement is T+1: money from any sale, including selling SGOV, can be spent only on the next
  trading day. Buy ONLY with settled cash, so there is never a good-faith violation.
- Fractional shares are allowed. Skip any order under $1.00.
- SGOV (0-3 month T-bill ETF) is money waiting to be invested, not a long-term holding.
- Settled cash available today: $____
- My risk level today: ____ (1-10). It sets the loss limit used below:
  1 = 1%, 2 = 2.5%, 3 = 5%, 4 = 10%, 5 = 15%, 6 = 20%, 7 = 30%, 8 = 40%, 9 = 50%,
  10 = no stress-loss limit (only the per-holding caps in rule 3 apply).
  If I leave it blank, ask me; do not assume one.

## My holdings (shares as of 2026-10-23)
SGOV 31.17 | VTI 2.52 | SCHD 23.04 | AVUV 2.17 | VXUS 3.14 | SPYG 1.44 | SOXL 0.774368
QQQM 0.215404 | SOXX 0.040852 | OKLO 0.272778 | ARR 0.125027 | TQQQ 0.022524 | JEPI 0.017503

Prices: use today's closing (or latest) prices. If you can browse, look them up and list each one
with its time. If you cannot, STOP and ask me for them. Never estimate a price.

## My current recurring buys (about $27.41 per trading day)
VTI $10.45/day, SCHD $5.97/day, VXUS $2.98/day, AVUV $2.98/day, QQQM $2.98/day, SPYG $2.05/day,
JEPI $1.00/week. Keep the same daily total, but you may move it between tickers.

## Method
1. Stress loss = sum over holdings of (weight x that holding's worst historical drop from peak).
   It is the loss if every holding repeated its worst drop at the same time. HARD RULE:
   stress loss <= my loss limit, measured on the whole portfolio value (cash included).
2. Worst drops to use (a longer-lived fund on the same index stands in for a young one):
   SGOV 0%, SCHD 33%, VXUS 36%, JEPI 50% (assumed), VTI 55%, AVUV 62% (via VBR), SPYG 68%,
   SOXX 70%, ARR 80%, TQQQ 82%, QQQM 83% (via QQQ), OKLO 85%, SOXL 90%.
   For any other ticker, use its worst drop since inception, and at least 50% if it is under
   10 years old.
3. No single holding may use more than 25% of the loss budget:
   weight x worst drop <= 0.25 x limit (skip this at level 10). No holding above 35% of the portfolio (SGOV excepted).
4. Expected return above cash = volatility x Sharpe, where Sharpe is a blend: 0.3 for every
   asset, moved toward the asset's own 10-year Sharpe in proportion to its years of history
   (at most halfway, for 10+ years). Then rank by expected GROWTH:
   growth = expected return - 0.5 x volatility^2. This charges leveraged and very volatile
   names (SOXL, TQQQ, OKLO) for volatility drag.
5. Universe: my holdings plus these ETFs: VOO, QQQ, VTV, VIG, USMV, QUAL, MTUM, RSP, IJR, VEA,
   VWO, XLK, XLV, XLF, XLE, XLU, XLP, XLI, VNQ, GLD, DBC, BIL, SHY, IEF, TLT, TIP, BND, LQD, HYG,
   UPRO. You may also suggest up to 5 S&P 500 stocks or disruptive-theme / small-cap names, but
   only with a stated worst drop and growth estimate. Skip anything with under 3 years of prices,
   and skip a fund that largely duplicates one already chosen (correlation above ~0.85).
6. Spend the loss budget on the best growth per unit of worst drop, subject to rules 1 and 3,
   keeping correlation in mind. Everything not invested stays in SGOV.
7. Optional tilt: if I paste ratings (Buy / Overweight / Hold / Underweight / Sell), add
   +0.30 / +0.15 / 0 / -0.15 / -0.30 to that asset's Sharpe. A tilt can move money between
   assets but can NEVER raise the stress loss above the limit.
8. Do not churn: skip any trade smaller than 0.5% of the portfolio.
```

```
## What to output, in this order
1. TODAY'S ORDERS: SELL and BUY orders in dollars, sells first, one-line reason each. T+1:
   - Today's buys <= today's settled cash.
   - Proceeds from trims settle tomorrow and buy tomorrow; what is not needed goes into SGOV.
   - For new money, SELL SGOV today at my daily pace (~$27.41) so it has settled by tomorrow.
   Then list TOMORROW'S BUYS, paid for by today's sales.
2. TARGET MIX: a table with ticker, price, current $, current %, target %, trade $, worst drop,
   weight x worst drop, and growth above cash. Show the total stress loss now and at target.
   Below it, ONE sentence per fund naming the number that set its weight (growth above cash,
   worst drop, the cap that bound it, or the fund it duplicates). Then name the 3 best funds you
   did NOT choose, and why.
3. NEW DAILY BUYS: split ~$27.41/day across the tickers under target, in proportion to how far
   under they are. If none are under target, say "pause recurring buys".
4. WHAT THE LEVEL COSTS: for risk levels 1, 3, 5, 7 and 10, the approximate % invested and
   expected growth a year. For reference, my app's walk-forward backtest (2012-2026):
   1% limit -> 1.9%/yr, worst drop 0.8%; 5% -> 3.4%/yr, 3.9%; 10% -> 5.3%/yr, 8.3%;
   VTI alone -> 14.6%/yr, worst drop 35%.
5. CAVEATS, in two lines: what the estimate cannot know, and that a future crash can be deeper
   than any past one.

## Rules for you
- Never invent a price, a worst drop or a statistic. If unsure, say so or ask me.
- Never call the loss limit a guarantee: it bounds a repeat of history, nothing more.
- Never change my risk level. If something I ask conflicts with it (for example "maximize
  windfall gains"), show the trade-off in numbers and let me pick a level.
```